

Automatizing the search for mass resonances using BumpNet, 13th LHCP Poster Session - 6th May 2025

Jean-François Arguin¹, Émile Baril¹, Ilan Bessudo², Maryna Borysova², Shikma Bressler², Samuel Calvet³, Shalini Epari¹, Ethan James Meszaros¹, Elad Kliger², Michael Kwok Lam Chu², Hoang Dai Nghia Nguyen¹, Julien Noce Donini³, Eva Mayer³, Bruna Pascual³, Amit Shkuri², Muhammad Usman¹

¹Université de Montréal ²Weizmann Institute of Science ³Laboratoire de Physique de Clermont - UCA - CNRS/IN2P3

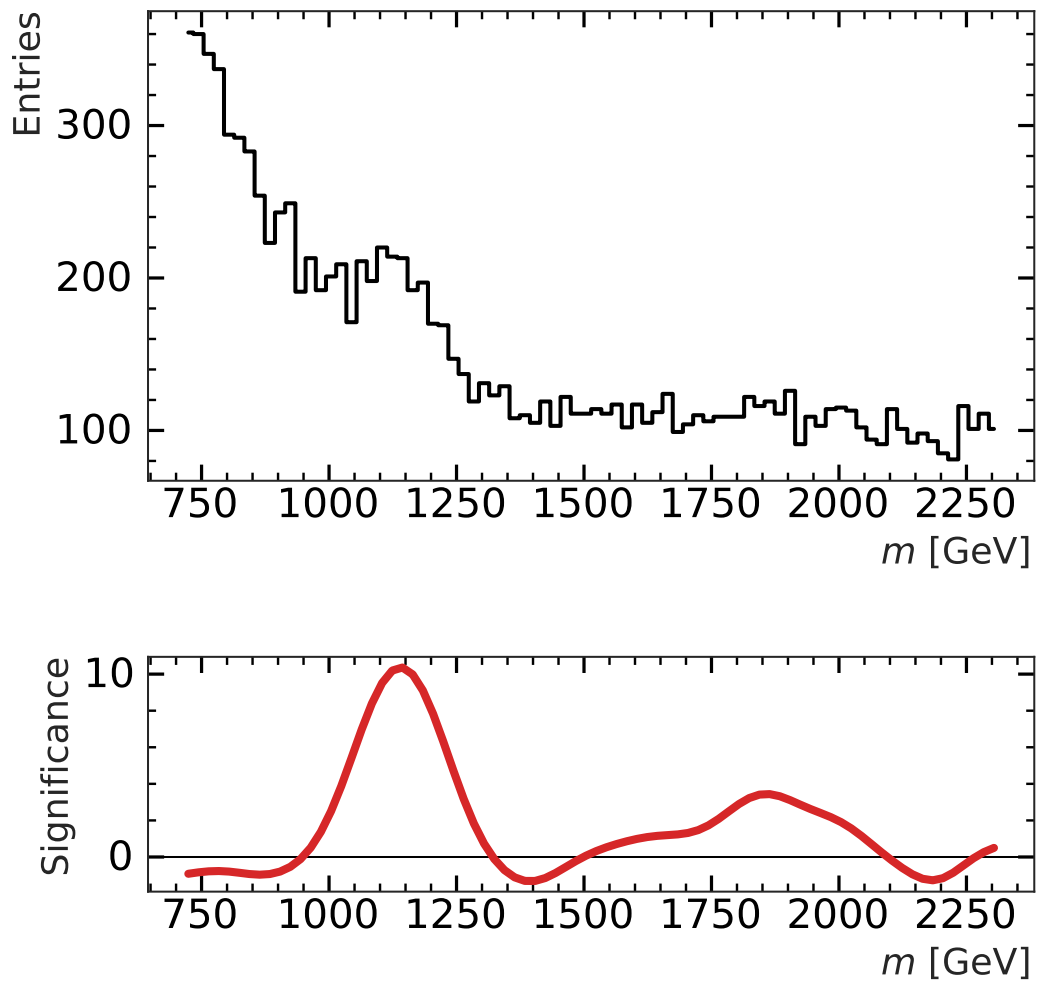


Motivation

- What?** Train a neural network to **identify mass bumps in real data** without the need of simulation or analytical fit to estimate the background
- Why?** Exploit the **discovery potential of the data**
 - Impossible to check all possible searches with the traditional analysis
 - Many possible resonances in unexplored final states

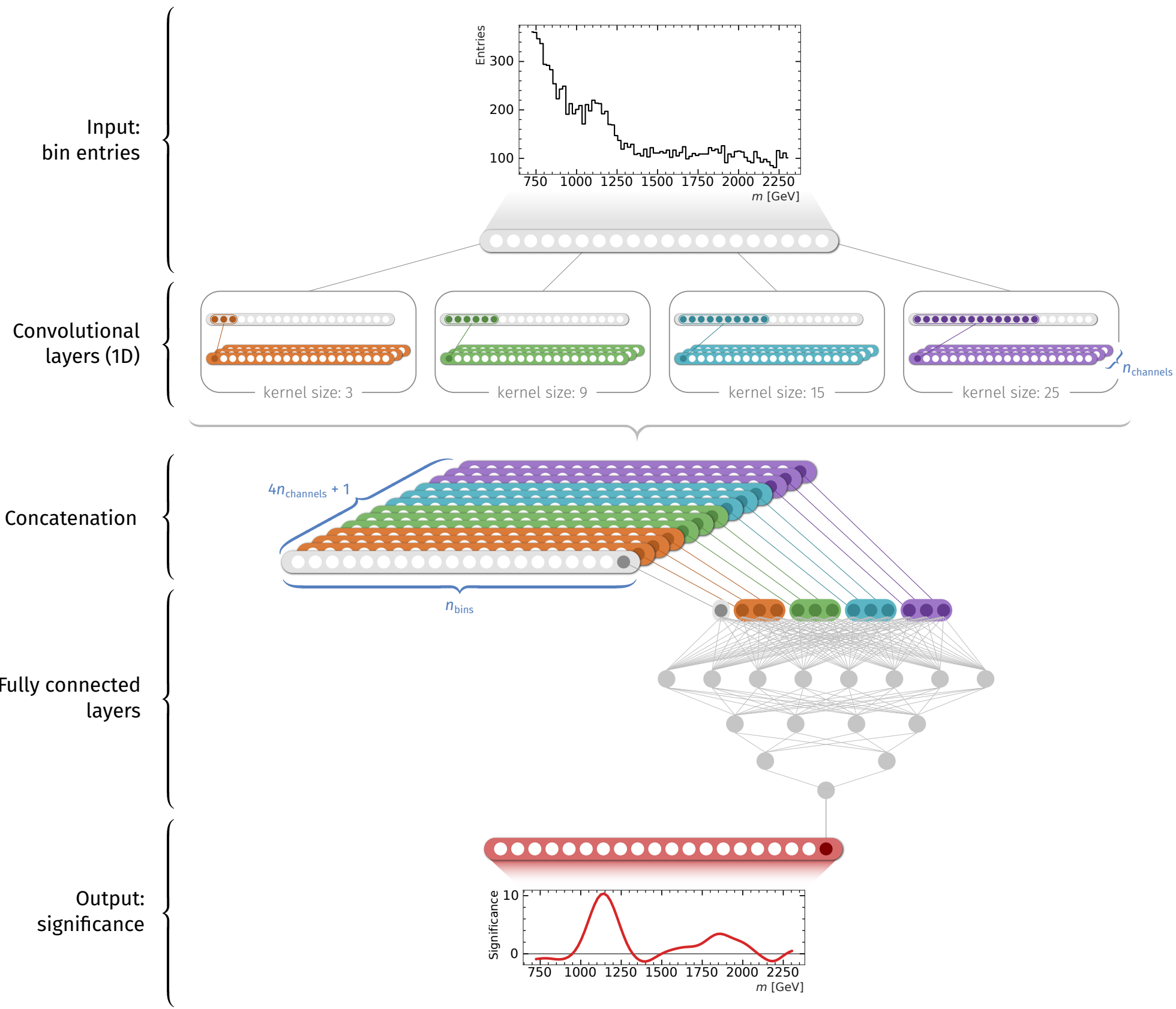
Data Directed Paradigm for bump searches

- The **Data Directed Paradigm (DDP)** [1] is a search strategy to efficiently identify regions of interest in the data. It requires two ingredients:
 - Property of the SM on which deviations can be searched for
 - Tool to scan the observable-space in search for deviations
- BumpNet** = NN mapping invariant mass to statistical significance for bumps [2]
- Network **mapping invariant mass distributions to statistical significance**
 - Input:** bin entries of the invariant mass distribution
 - Target:** bin-by-bin statistical significance of having a signal with respect to a smooth background, calculated using the **profile likelihood ratio** hypothesis test



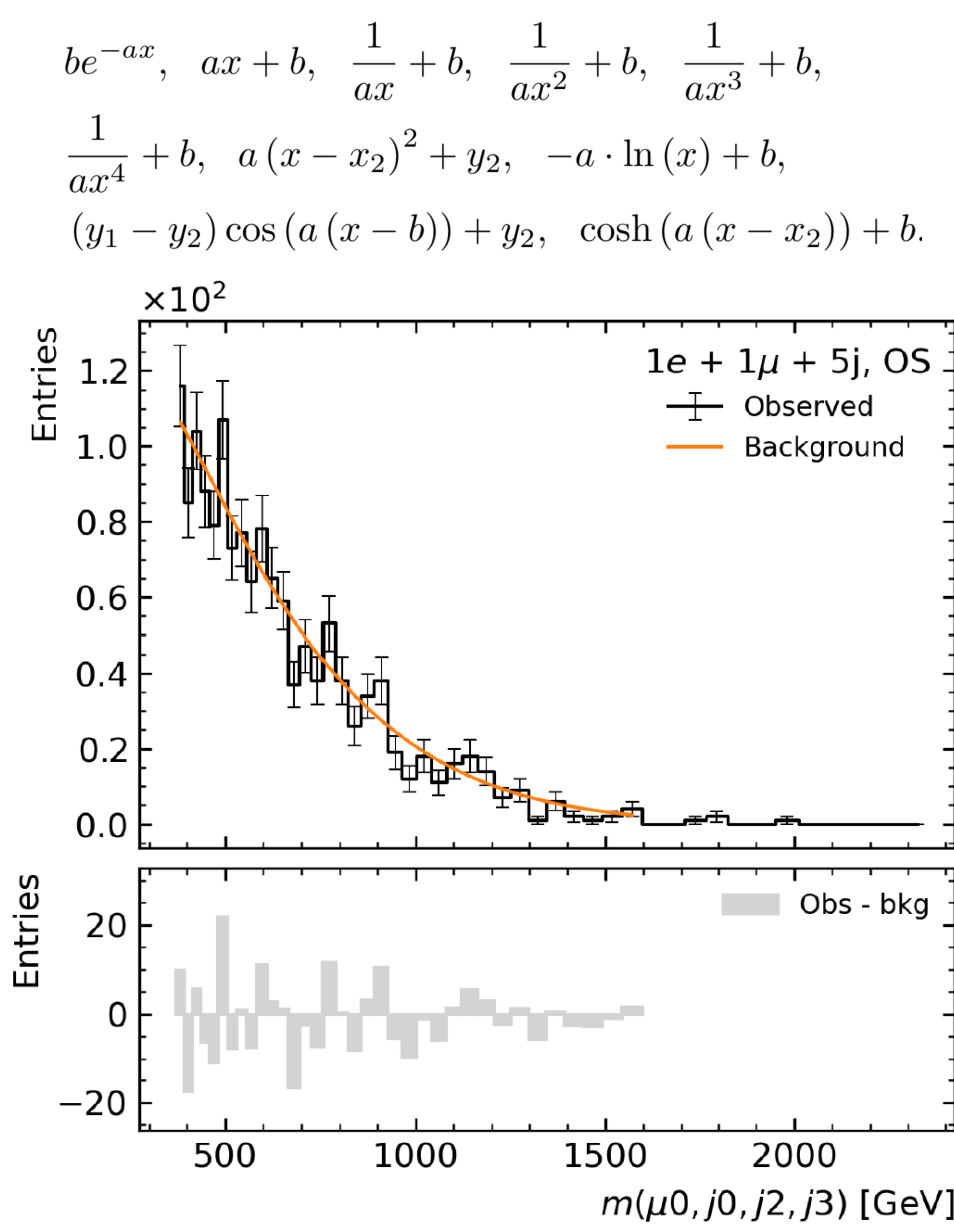
Neural network architecture

- Use of 1D convolution layers followed by a dense layer
- Intuitive and agnostic to the number of bins in the histogram

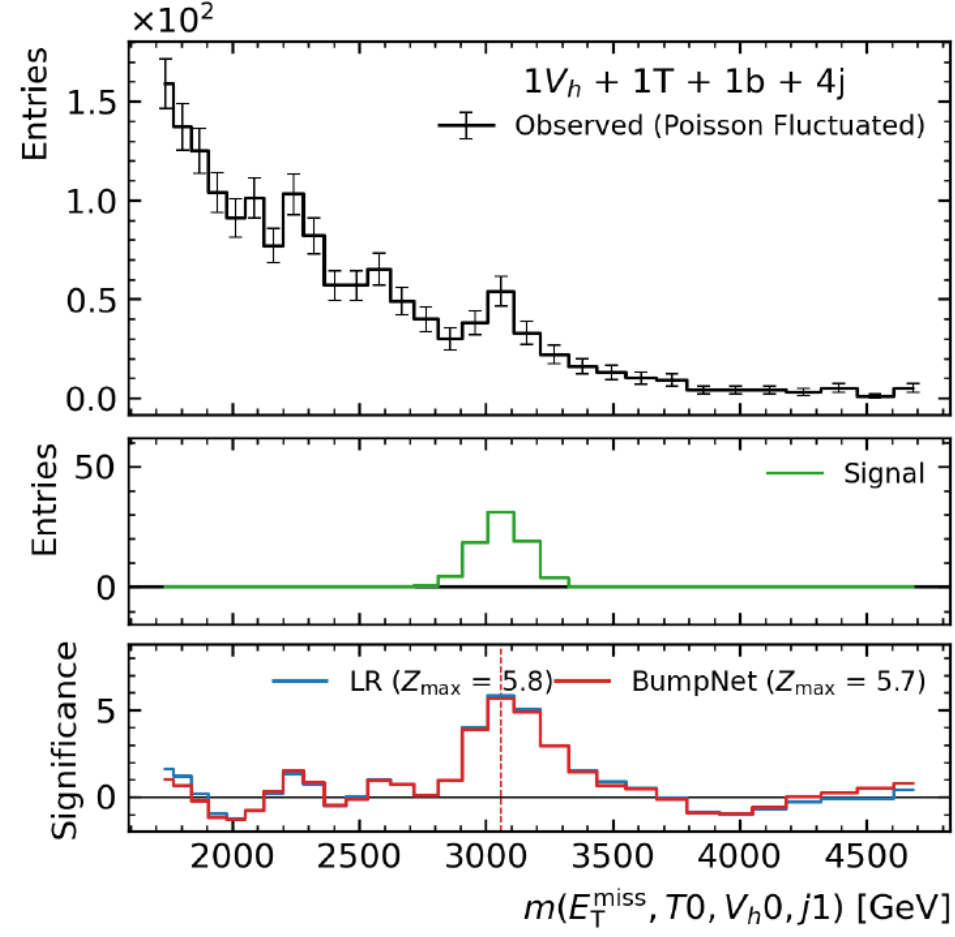


Training data

- Synthetic data generated from **injecting a Gaussian signal on two types of backgrounds**: analytical functions and fits to simulation data



- Poisson fluctuation allows for a more realistic distribution



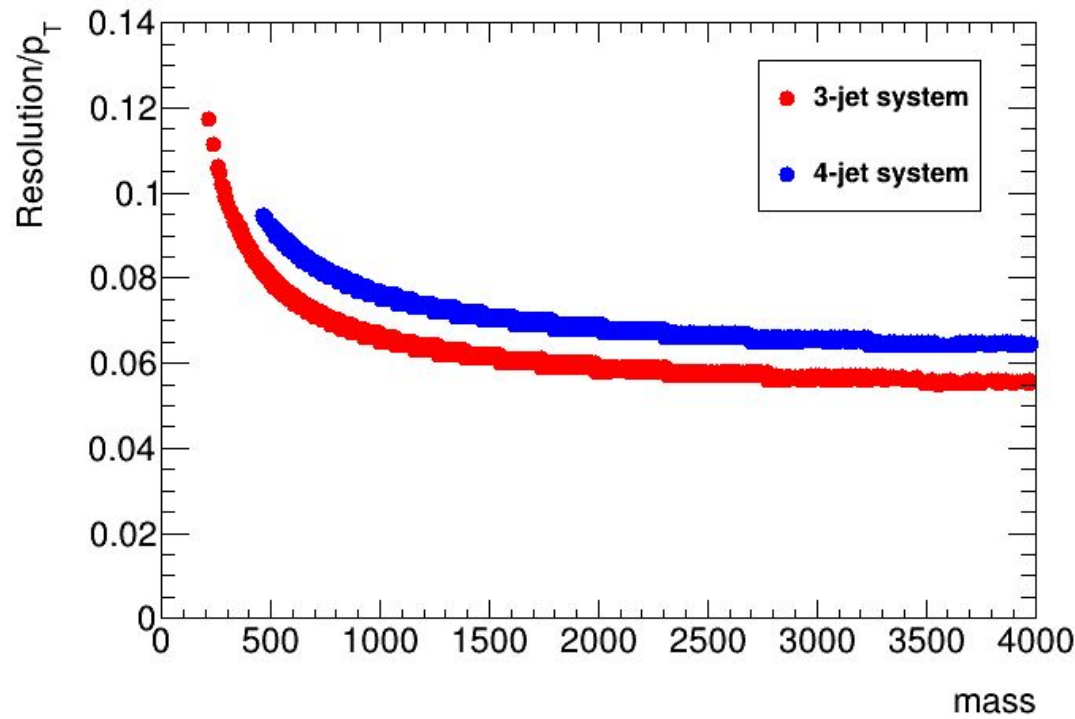
Acknowledgements

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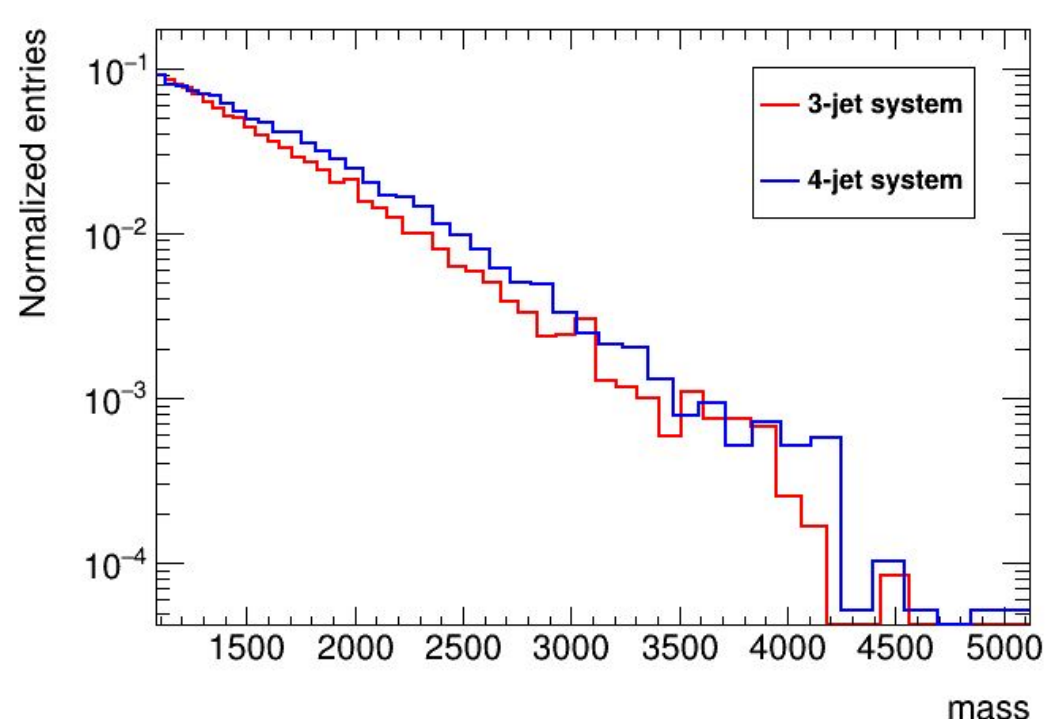


Histogram processing and calibration

- Using the **Dark Machines dataset** [3]
 - Designed to test anomaly detection techniques
 - Dataset equivalent to 10 fb⁻¹ with highest cross-section processes at the LHC
- Mass histograms with **all possible combinations** of the following objects:
 - Electron Photon Reconstructed leptonic Z Boosted top
 - Muon Jet Boosted hadronic W/Z High mass jet (m > 200 GeV)
- Additional kinematic cuts on MET and pT of leading objects
- Split the data according to jet multiplicity to improve S/B and reduce the look-elsewhere effect
- Total of 30 000 mass histograms
- Rebinning that reflects the detector resolution**, using $p_T \approx m/2$



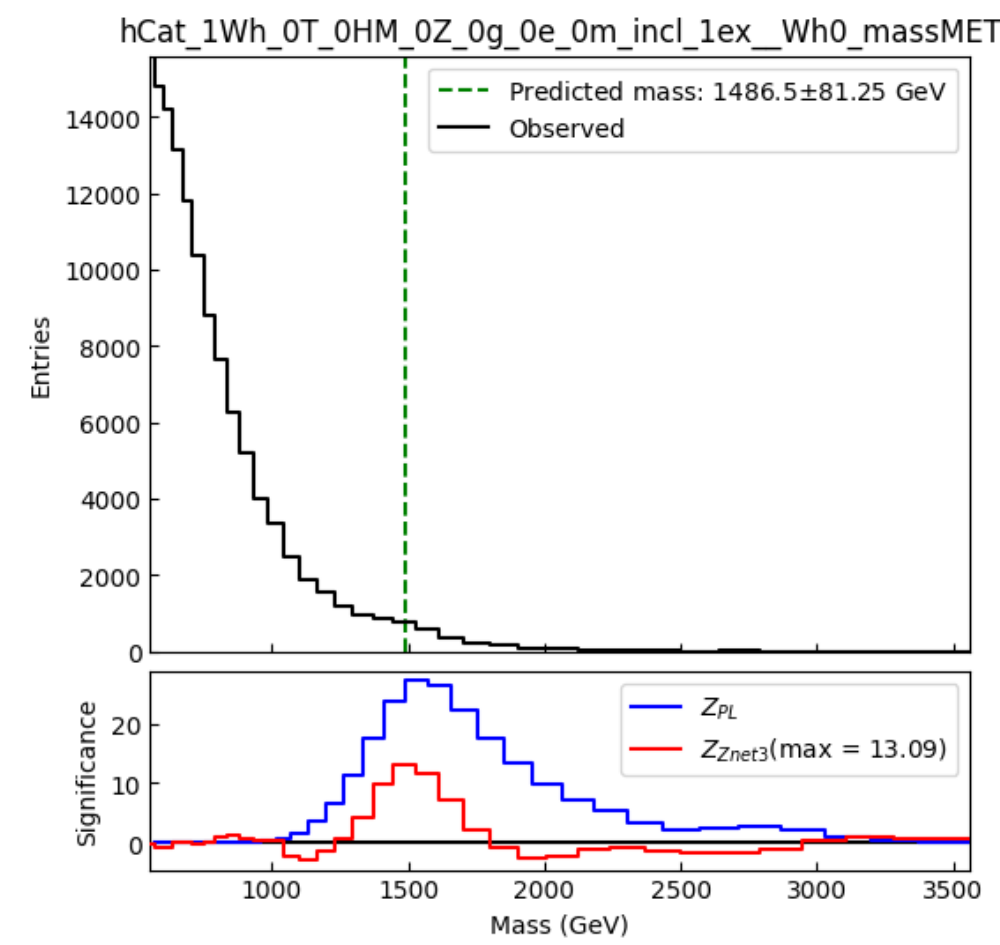
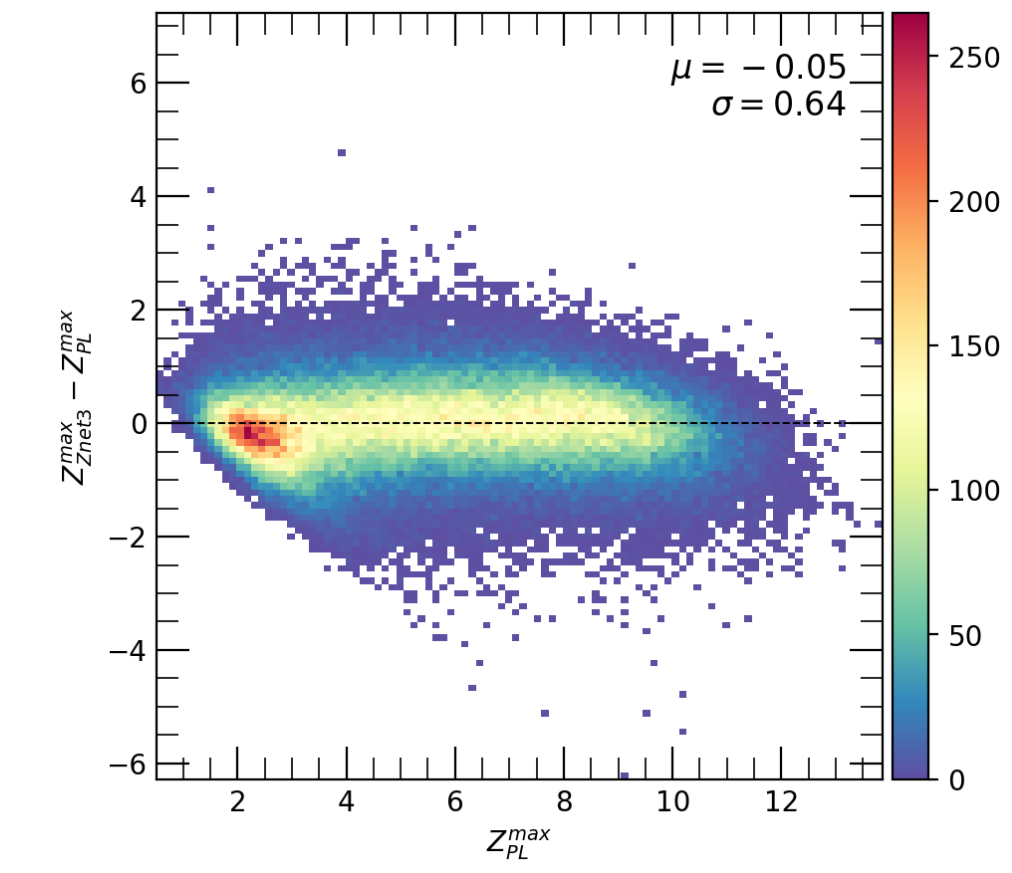
Resolution is higher for m(4j) than m(3j), also larger for small masses



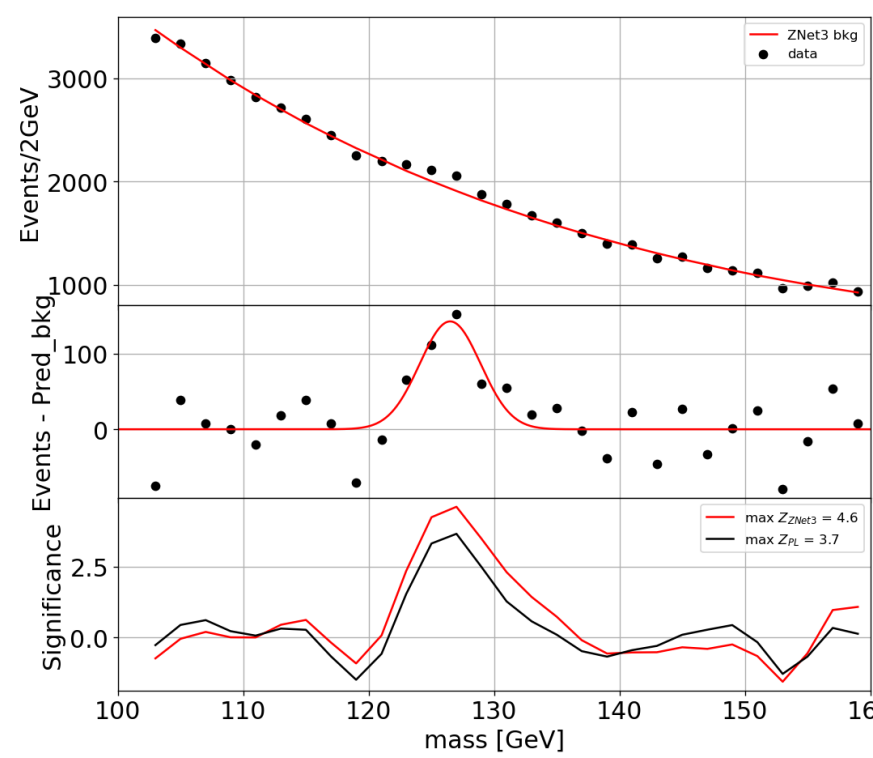
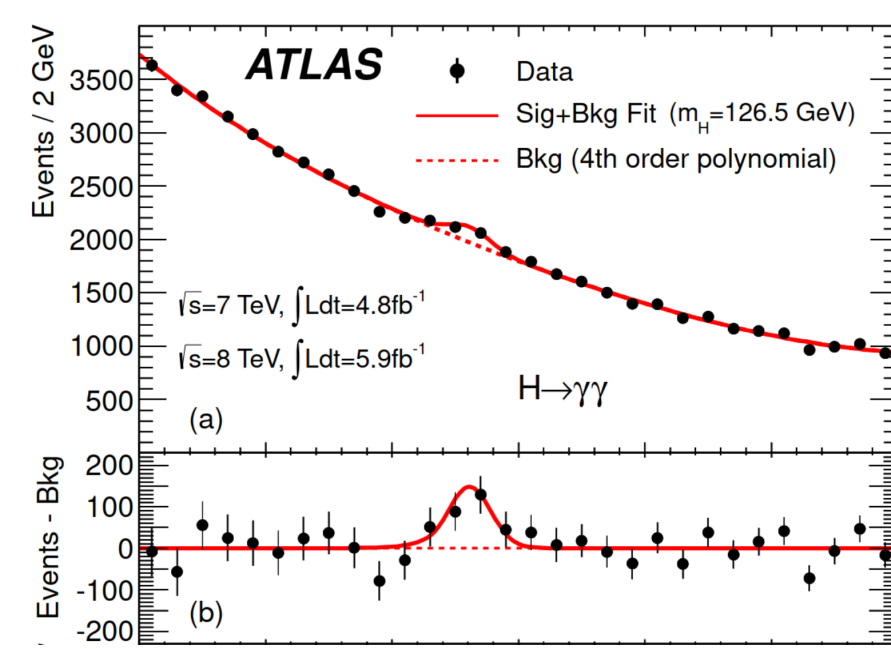
Binning reflects this with larger bin width when resolution is smaller

Performance and finding BSM signals

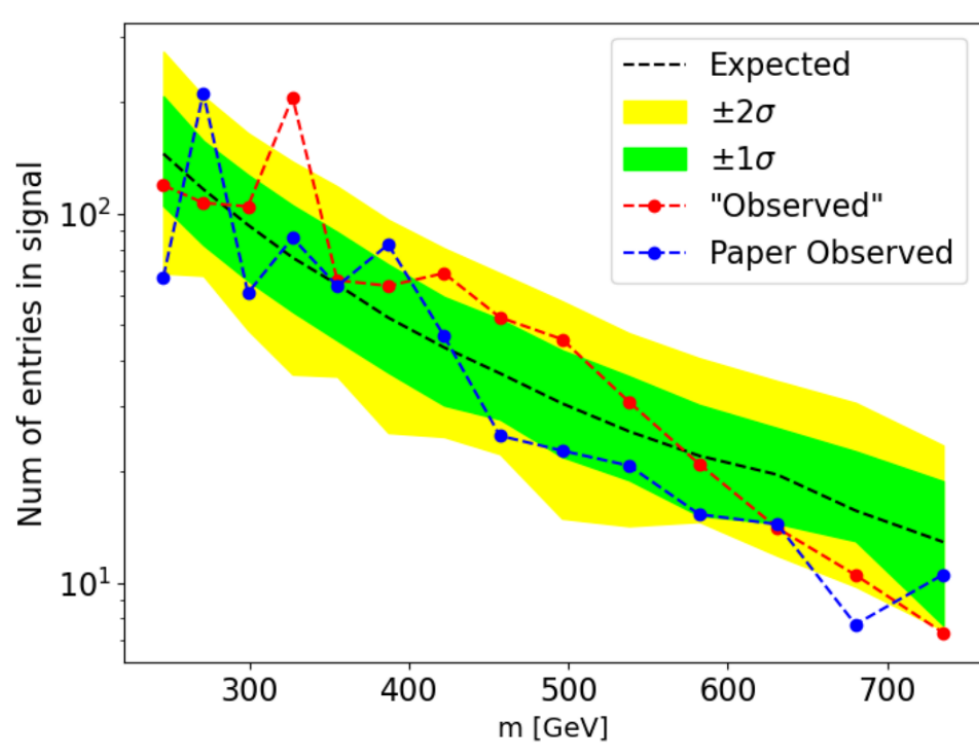
- Accurately predicts maximum significance** with no bias and a variance of ± 0.64
- Successfully finds BSM signals**



- Promising results when **finding the Higgs bump**
 - Data sampled from the plots using a digitizer
 - Predicted significance of 4.6σ whereas the ATLAS significance is 3.7σ [4]



- Ongoing ATLAS analysis to use BumpNet over Run 2 data
- Developing statistical analysis to **set limits on the number of signal entries** for an observed Z
- Currently testing if the network can find **more BSM signals**
 - $Z' \rightarrow \ell\ell$
 - MSSM $H \rightarrow \ell\ell$
 - $W' \rightarrow \ell\nu qq$
 - $Z + W'/Z' \rightarrow \ell\nu qq$
 - $LQ \rightarrow \ell q$
 - 2HDM VLL
 - SUSY RPV $\rightarrow \ell + jets$



- Developing a global significance statement based on bumps from statistically correlated histograms

References

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